B. Tech Summer / Winter - 2024
(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering **Sem:- V**
Subject: Design & Analysis of Algorithm **Subject Code: CS5T003**
Marks: 100 **Date of Exam: 02/12/2024** **Time: 3 Hrs.**

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Recall basic concepts of algorithm in analysis and Design of algorithms.
2. Examine Recurrence relations, solutions of recurrence of searching sorting methods.
3. Apply Greedy methods used for analysis and Design of Algorithm.
4. Analyse Dynamic Programming concepts in designing algorithm.
5. Evaluate advanced techniques and tools available for algorithm analysis and development.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Algorithm with the characteristics of an algorithm.	L1/CO1	2
Q.2	State the term Asymptotic Notation. List out the types of Asymptotic notations.	L1/CO1	2
Q.3	List out the different algorithm design strategies.	L1/CO1	2
Q.4	State how Divide and Conquer Algorithm Works?	L2/CO2	2
Q.5	Rewrite relation between IPL and EPL.	L2/CO2	2
Q.6	Define Spanning Tree. Classify the algorithms used to implement *Minimum Cost Spanning Tree.*	L1/CO3	2
Q.7	Restate Principle of Job Sequencing with Deadline.	L2/CO3	2
Q.8	Write the concept of Knapsack Problem.	L3/CO3	2
Q.9	State the principle of Backward and Forward Multistage Graph.	L1/CO4	2
Q.10	Compare Partial Knapsack and 0/1 Knapsack.	L2/CO4	2
Q.11	Define Graph Coloring with example.	L2/CO5	2
Q.12	Give two solutions for 4-Queen Problem.	L1/CO5	2

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Explain in detail the growth rate of function.	L2/CO1	4
Q.14	Draw split & merge tree for array of 10 elements.	L3/CO2	4
Q.15	Write the algorithm for Binary Search.	L3/CO2	4
Q.16	Discuss the characteristics of Greedy Algorithm.	L2/CO3	4
Q.17	Differentiate between Dynamic Programming and Divide & Conquer Strategy.	L4/CO4	4
Q.18	Describe the concept of Backtracking with example.	L2/CO5	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Solve the following equation by using characteristic equation.	L3/CO1	7

$t_n = n$. if $n = 0$, $n = 1$, $n = 2$

$= 5t_{n-1} - 8t_{n-2} + 4t_{n-3}$. otherwise.

Q.20	Explain the Amortized Analysis & different methods used in Amortized analysis.	L2/CO1	7
Q.21	Identify the best possible sequence for the job with deadline.	L2/CO3	7

Jobs	1	2	3	4	5	6	7
Profit	3	5	18	20	1	6	30
Deadline	1	3	4	3	2	1	2

Q.22	Determine optimal solution for Knapsack Problem. Using greedy strategy for following instances n=7, m=15.	L3/CO3	7

Object	1	2	3	4	5	6	7
Weight	2	3	5	7	1	4	1
Profit	10	5	15	7	6	18	3

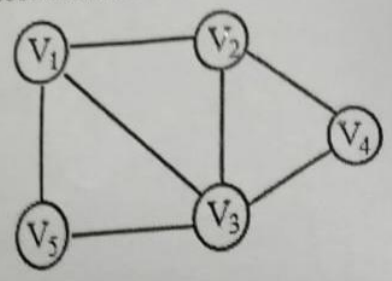
Q.23	Write an algorithm to generate LCS. Apply algorithm for following string & generate LCS.	L3/CO4	7

X = POLYNOMIAL

Y = EXPONENTIAL

Q.24	Perform Graph Coloring on the given graph. State the total number of	L5/CO5	7

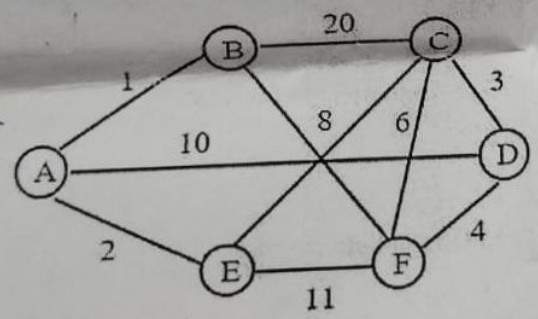
possible solutions.



SECTION D

1. **Solve any TWO Questions.**

Q. No.	*Questions*	Level/CO Mapped	Marks
Q.25	Consider the array given below to perform binary search. Find the average number of successful and unsuccessful search. Also find internal & external path length. a = [-15, -6, 0, 7, 9, 23, 54, 82, 101, 112, 125, 131, 142, 151] Also give the complexity of binary search.	L1/CO2	12
Q.26	Explain Minimum Cost Spanning tree. Find the minimum cost spanning tree using Kruskal algorithm and Prims algorithm for the graph given below :	L6/CO3	12



Q.27	Explain the following terms: a) Non-deterministic algorithm. b) Decision and optimized problem. c) NP-Hard problem. d) NP-Complete problem.	L6/CO5	12